

Unit 7, Lesson 5: Graphing Lines for Real-World Contexts

Benchmark

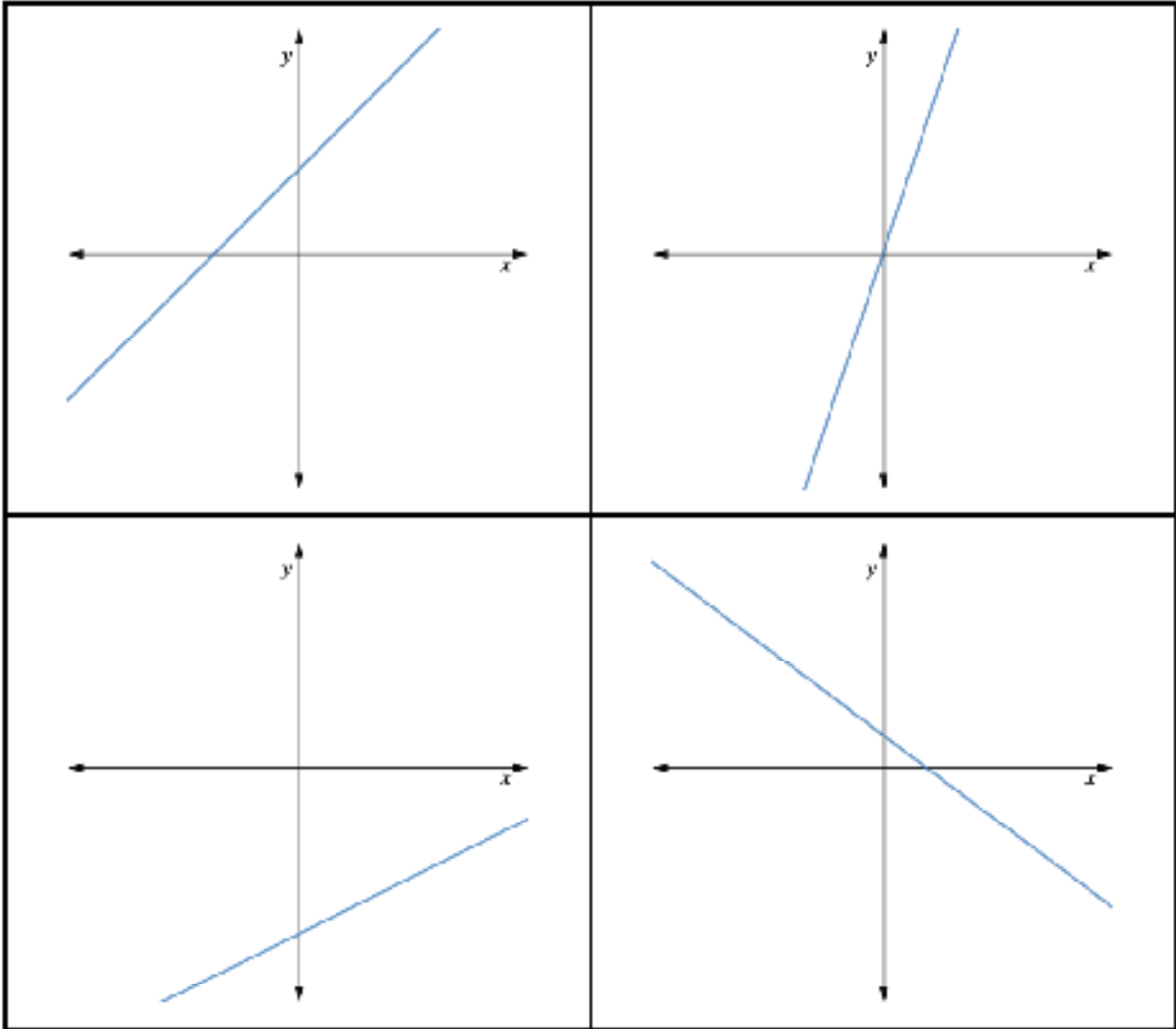
MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

Learning Target

- ☐ I can graph a two-variable linear equation from an equation, table, or written description given a real-world context.

7.5.1 Warm-Up: Which One Doesn't Belong?

The graphs of four linear relationships are shown.



1. Circle the graph that doesn't belong. Explain your reasoning.

7.5.2 Exploration: Graphing Linear Relationships for Real-World Contexts

1. When plugged into a USB charger, cell phones charge at a constant rate until they get closer to being fully charged.

When Michaela plugged in her phone, the battery was at 16%. After charging for 15 minutes, the battery level was at 41%.

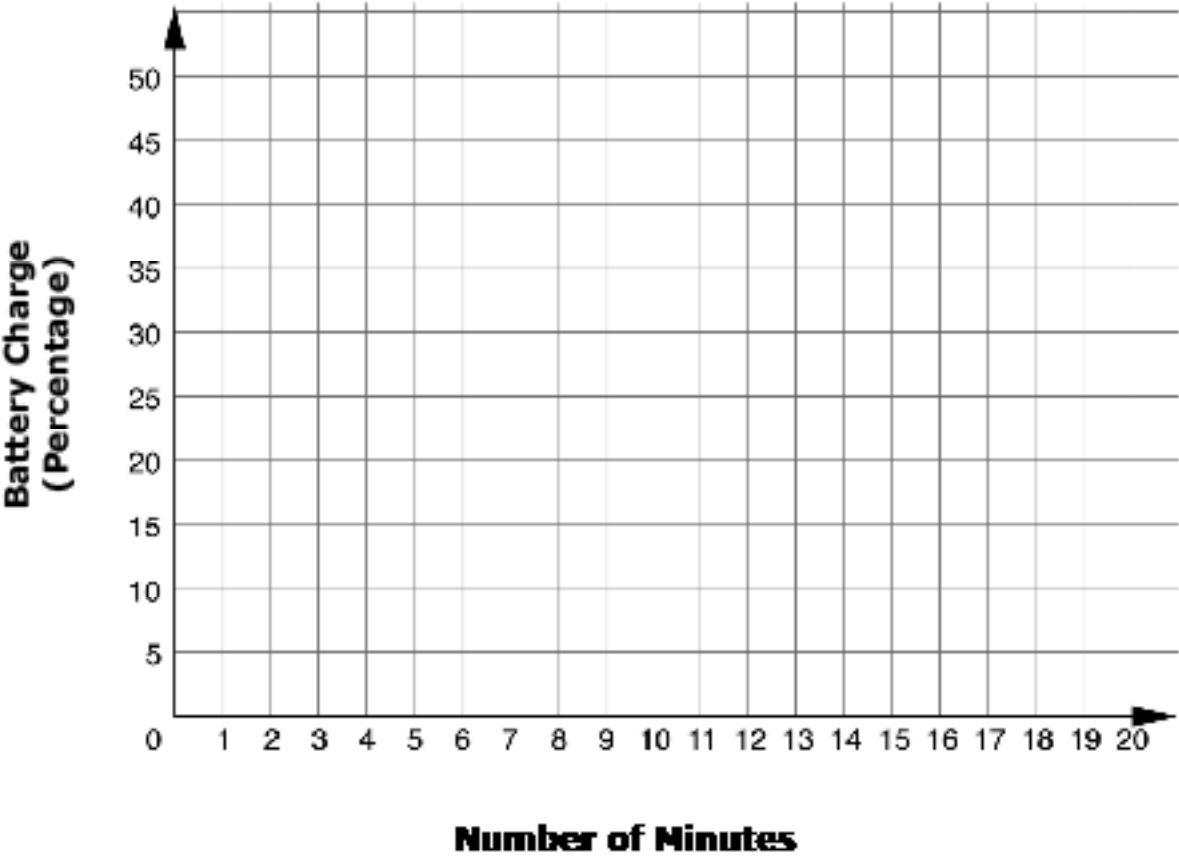
This linear relationship can be represented by a graph where x represents the number of minutes the phone was plugged in, and y represents the percentage of the battery charged.

Harry said there is enough information from this context to find two points on the graph of this relationship.

- a. Work with your partner to determine the coordinates of the two points.

Which point represents the phone's charge when Michaela plugged it in?	Which point represents the phone's charge 15 minutes after Michaela plugged it in?
(__, __)	(__, __)

b. Graph the linear relationship.

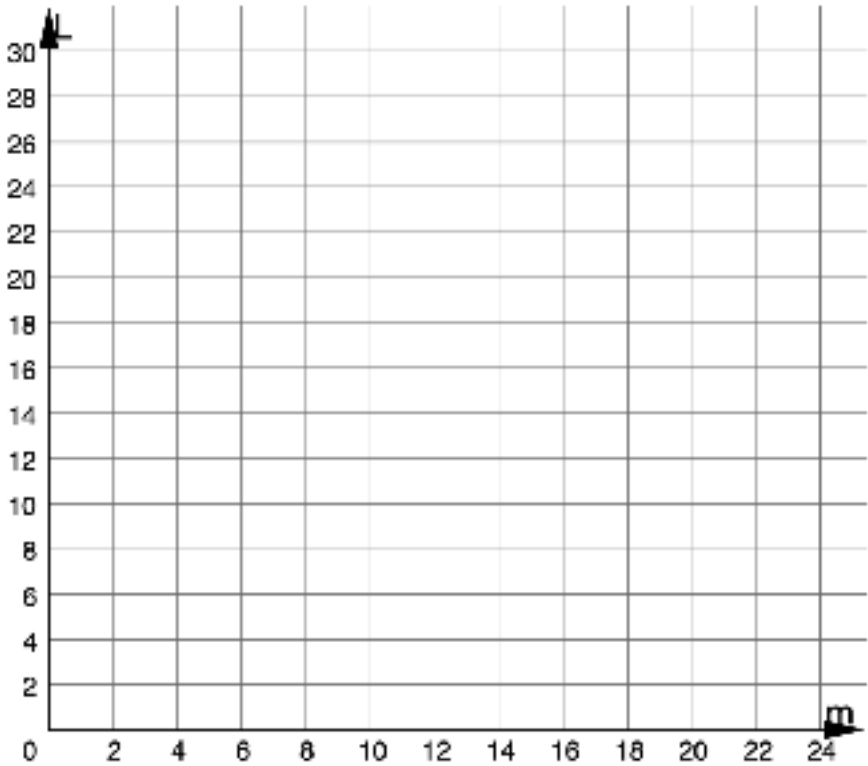


- c. What information was used to graph the relationship?
- d. Explain why the slope was not needed to graph this relationship.

2. The body length, L , of a milk snake in inches can be represented by the equation $L = 6.5 + 2m$, where m is the age of the snake in months for the first two years of its life.



a. Graph the relationship between L and m .



- b. What information was used to graph the relationship?
- c. Discuss with your partner how to determine from the equation which value represents the slope and which value represents the y -intercept. Write a brief summary of your discussion.

7.5.3 Bring it Together: Graphing Relationships from Real-World Contexts

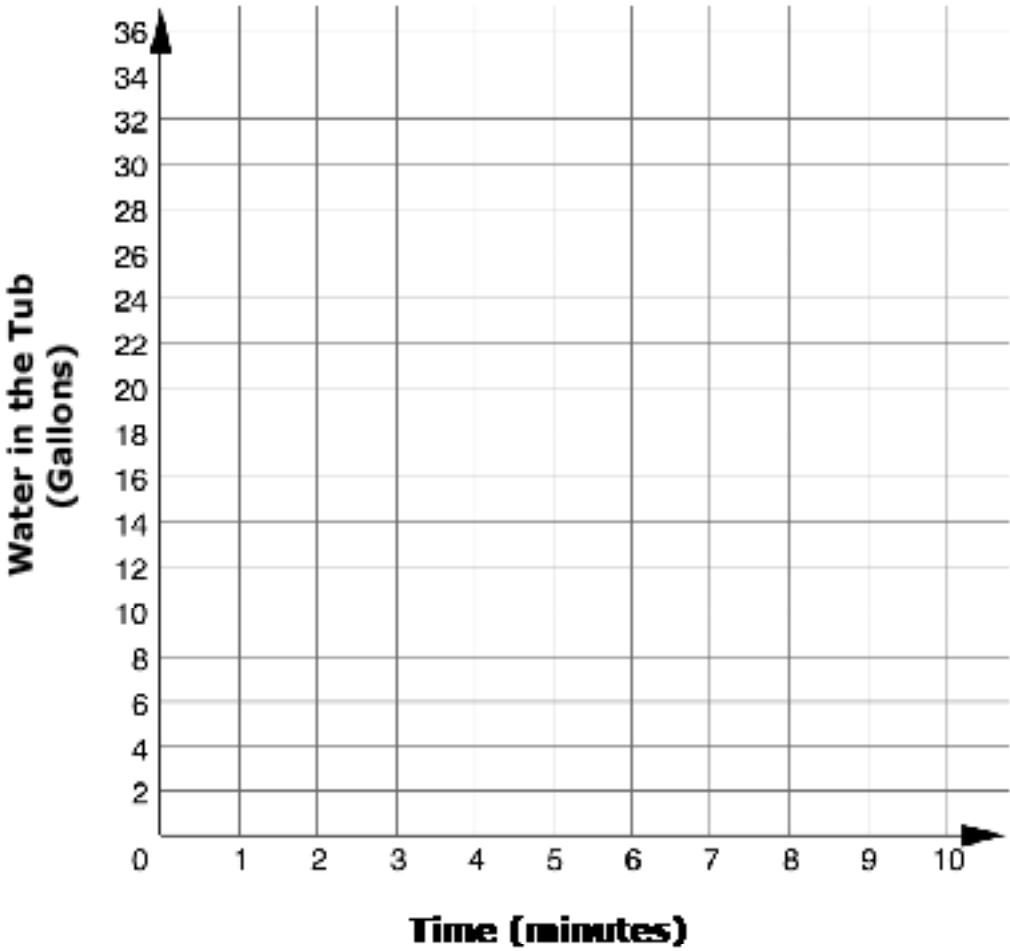
For each context, different information was used to graph the linear relationships.

To graph a linear relationship given a real-world context, the information that is necessary to know is either

- the slope only
 - the y -intercept only
 - the slope and y -intercept
- or
- two ordered pairs.
 - a table of values with five ordered pairs.

7.5.4 Guided Practice: Graphing Linear Relationships for Real-World Contexts

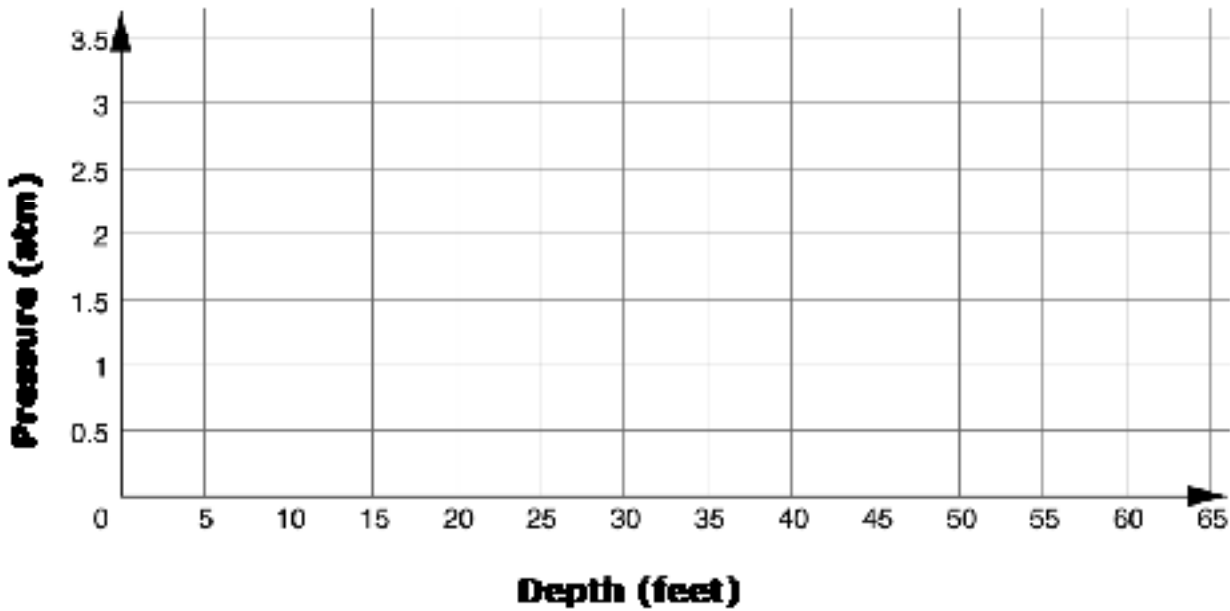
1. A bathtub drains at a constant rate. There are 30 gallons of water in the tub when Camille opens the drain. The bathtub drains at 12 gallons per minute.
- a. Which information about the relationship can be determined from the situation?
- the slope and y -intercept of the graph
 - at least two ordered pairs on the graph
- b. What are the slope and y -intercept, or at least two of the ordered pairs?
- c. Graph the linear relationship between time, in minutes, x , and the number of gallons in the bathtub, y .



2. Scuba divers wear protective equipment when diving because the pressure placed on the body increases as they dive, due to the weight of the water. At sea level, a diver experiences 1 atmosphere of pressure (atm). The table shows the pressure on a scuba diver at various diving depths.

Depth (feet)	0	20	40	60
Pressure (atm)	1	1.6	2.2	2.8

- a. Graph the linear relationship between depth in feet, x , and pressure, in number of atmospheres of pressure (atm), exerted on the diver, y .



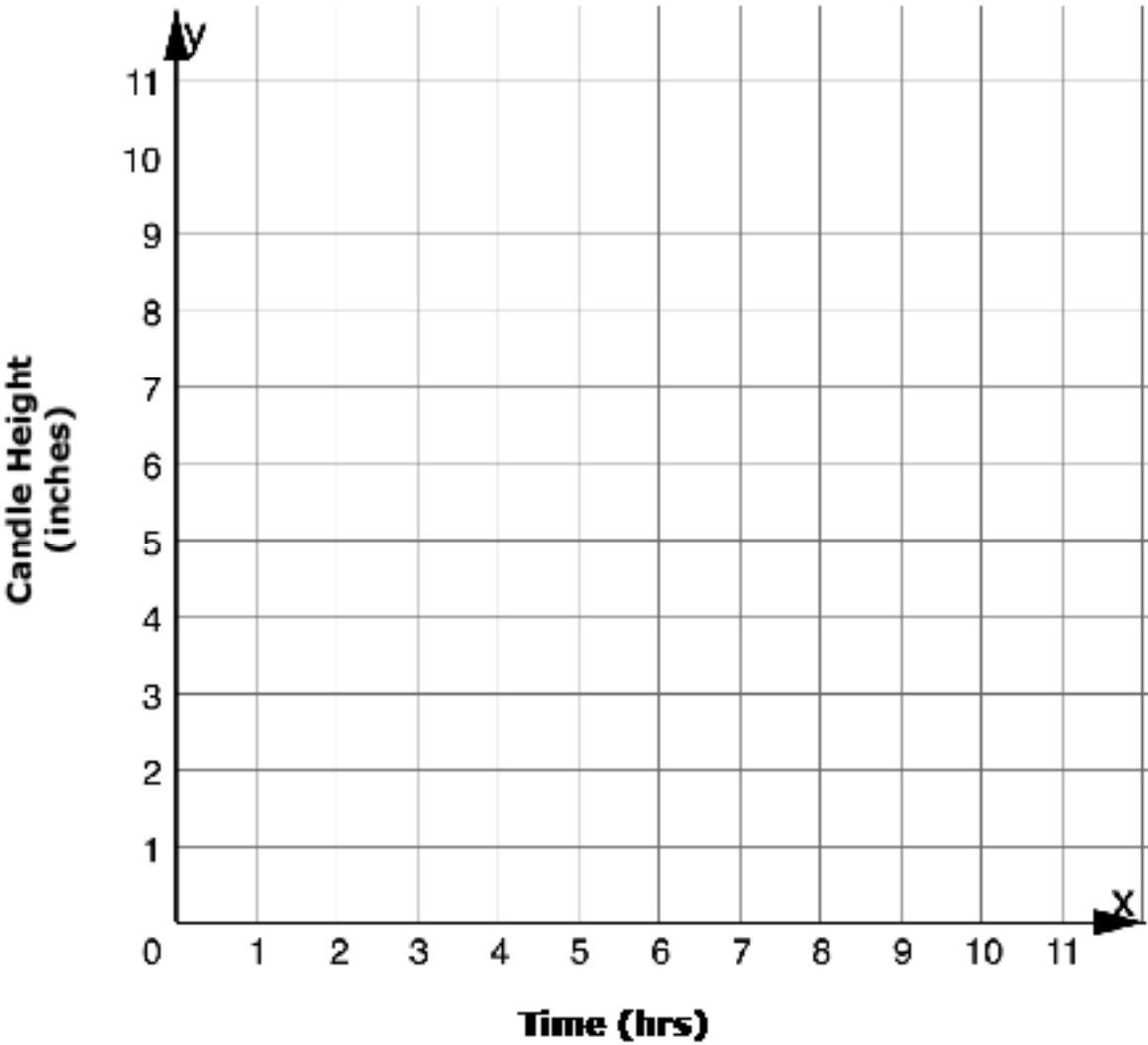
- b. What is the slope of the line?

7.5.5 Your Turn!

1. A candle is 10 inches long. Ximena lights the candle and records the length of the candle, y , in inches (in.) over x hours (hrs). The table shows Ximena’s recordings.

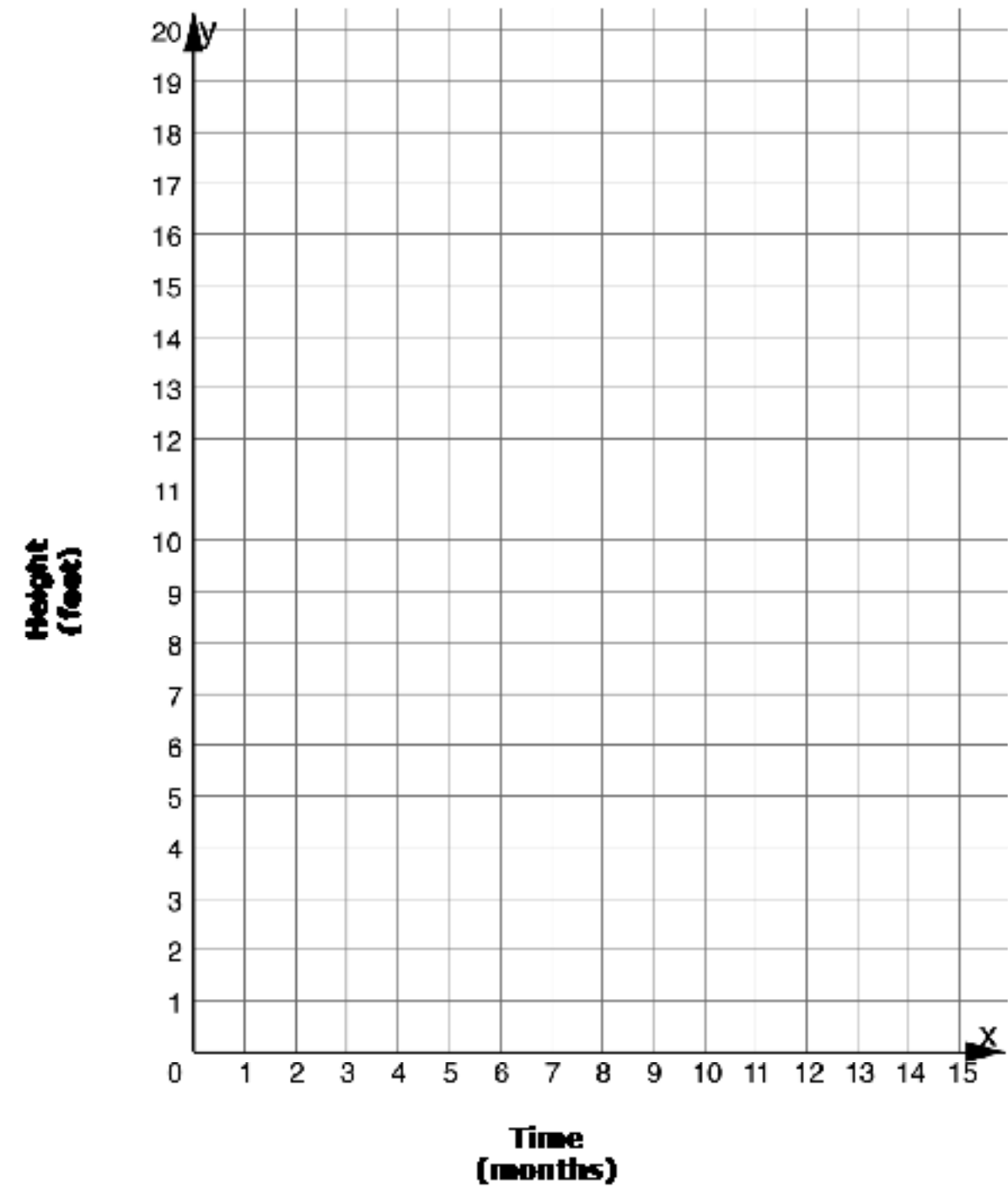
Time (hrs)	0	1	3	4
Candle height (in.)	10	8.75	6.25	5

Graph the relationship between candle height and time.



2. Gabriel planted a Leyland Cypress tree in his front yard. The tree was 11.5 feet tall when he planted it. It reached a height of 13 feet after 6 months.

Graph the linear relationship between the height of the tree in feet, y , and the number of months since it was planted in Gabriel’s yard, x .



7.5.6 Wrap-Up: Ticket Out the Door

The Johnstons turned their air conditioner off while they were out of town. When they returned home, the internal temperature of the house was 87°F . They turned the air conditioner back on and after 2 hours, the temperature had dropped to 83°F . The internal temperature of the house decreased at a constant rate until reaching 75°F .

1. Graph the linear relationship.

